



A QUICK ENGINEERING BREATH

THE PID LOOP, AT A GLANCE

Setpoint in. A controller reacts. The process responds. Feedback closes the loop.



the process, held constant across what follows

1
Process gain

10 s
Lag

1 s
Delay

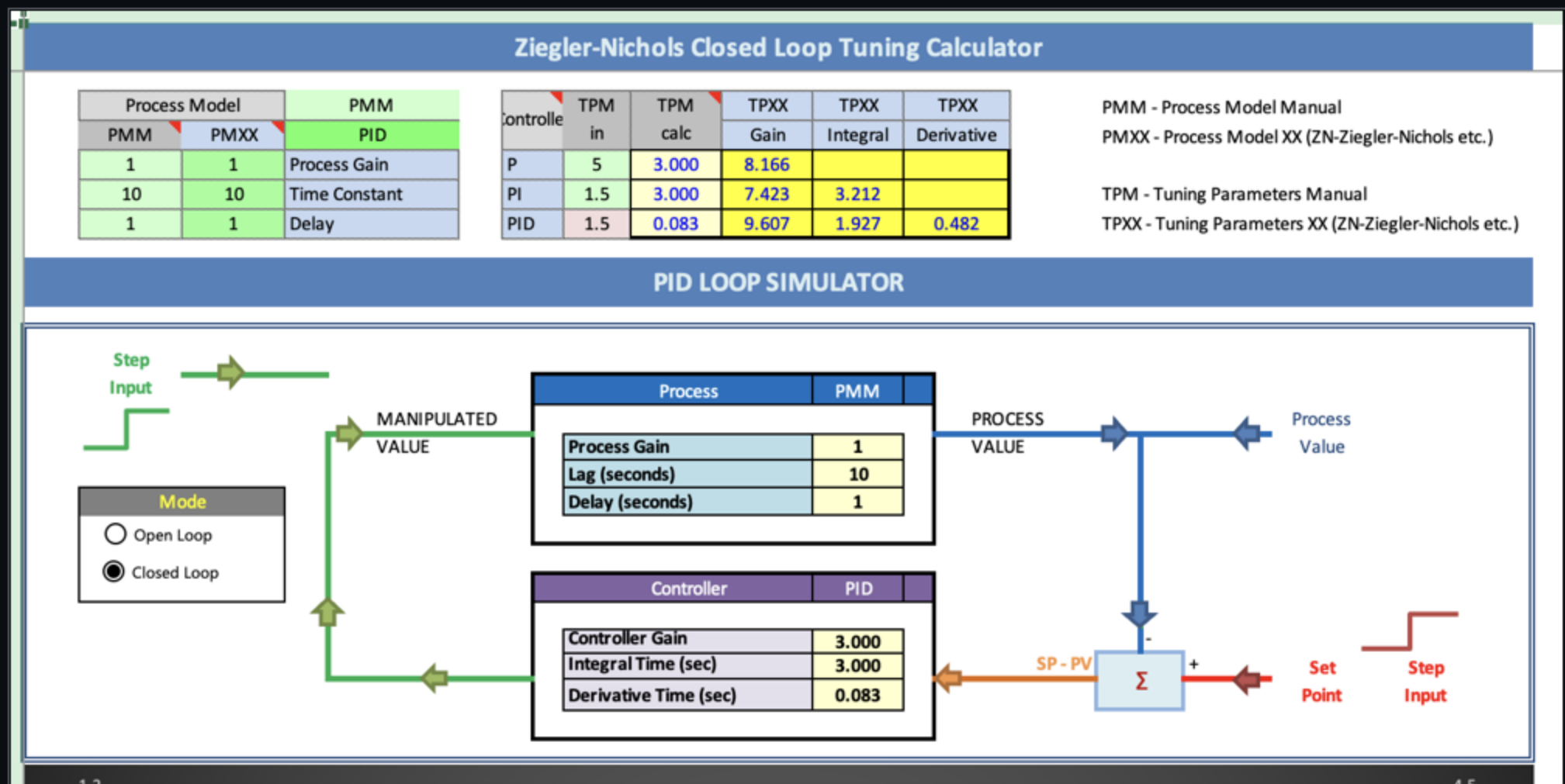
Same process, three tuning philosophies — next.



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THE SIMULATOR I BUILT THIS FOR

An interactive Ziegler-Nichols / PID loop simulator — the one behind the results that follow.

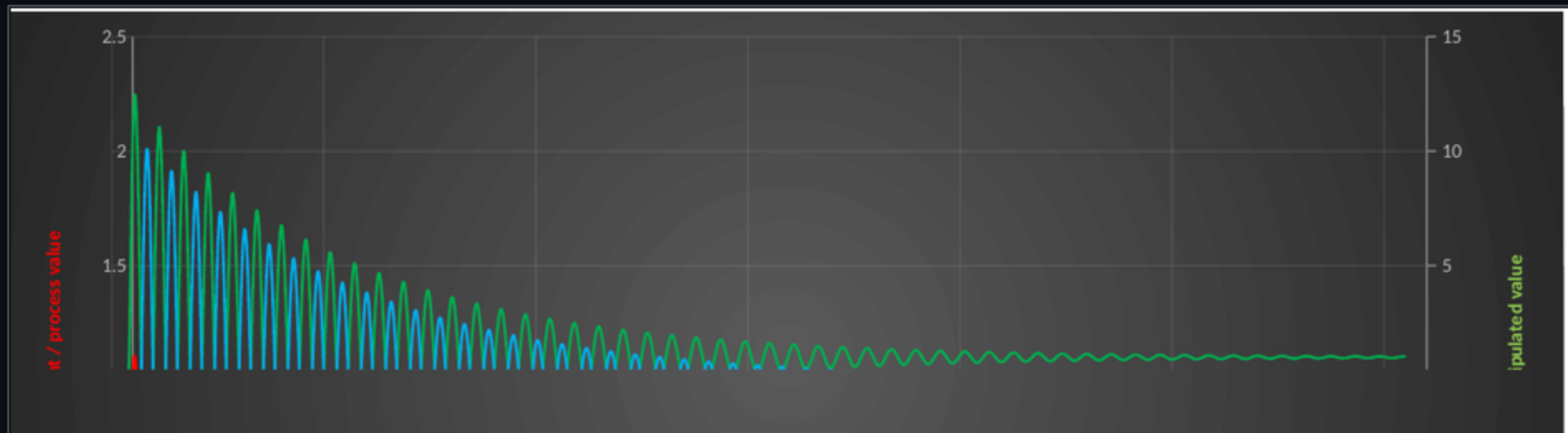


Same process, three tuning philosophies — next.



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CLASSIC ZIEGLER-NICHOLS



9.607

Gain

1.927

Integral (s)

0.482

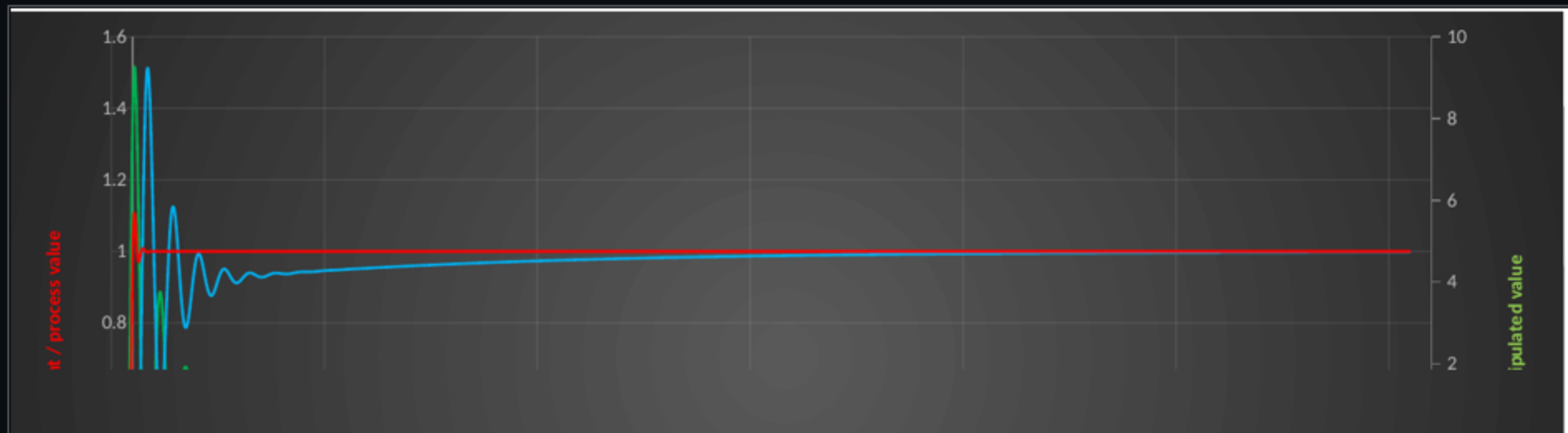
Derivative (s)

Aggressive — rings for a while before it settles.



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TYREUS- LUYBEN



7.423

Gain

8.480

Integral (s)

0.612

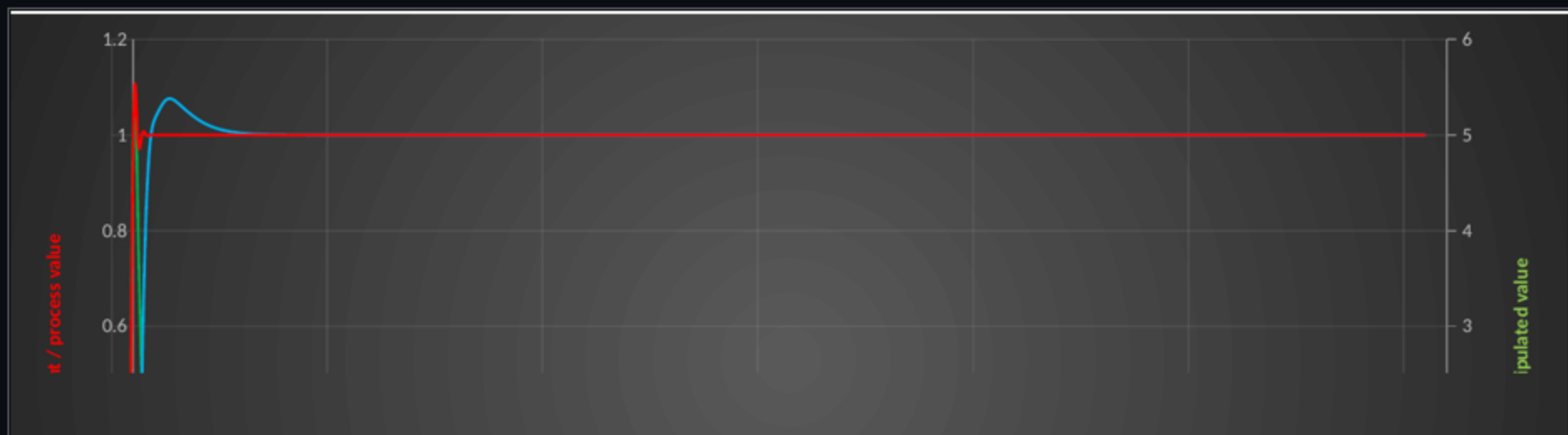
Derivative (s)

Balanced — a few oscillations, then settles fast.



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NO-OVERSHOOT VARIANT



3.266

Gain

1.927

Integral (s)

1.272

Derivative (s)

Conservative — smooth, no ringing at all.